RevIN Experiment 3: Centering and Transform Variants

1 Theoretical overview and goal

Mean centering assumes the look-back average is the appropriate location anchor for the future. Last-value centering instead predicts changes around the most recent observation, which may better suit local trends. The inverse hyperbolic sine transform

$$z = \operatorname{asinh}(x)$$

is approximately linear near zero and logarithmic for large magnitudes, giving a reversible compression of large standardized magnitudes after instance normalization.

Experimental goal. Test two lightweight changes to RevIN's inductive bias: a more local location anchor and a nonlinear magnitude compression as appendix ablations, without confounding either transformation with the training-loss choice.

2 Implementation details

The variants use non-affine instance normalization and are each trained with both MSE and normalized MSE:

Variant	Center	Transform	Losses
`instance`	Look-back mean	Identity	MSE, nMSE
`revin_last`	Last observation	Identity	MSE, nMSE
`revin_arcsinh`	Look-back mean	asinh	MSE, nMSE

The concrete method names append `_mse` or `_nmse`; for example, `revin_last_mse` and `revin_last_nmse`. The historical `revin_last` and `revin_arcsinh` labels denote non-affine instance normalization despite retaining the RevIN prefix. Every transformation is inverted before data-space evaluation. These appendix variants run in the full and ultra profiles, over the same nine datasets, five settings, PatchTST (plus DLinear in ultra), three seeds, optimizer, batch size, 10,000-step budget, and validation frequency. Configuration mapping is defined by `method_args` in `src/slurm/run_revin_experiment.sh`; layer behavior is in `src/utils/models.py`. The root front runs the separate training and table stages in order and resumes from signature-matched completion markers.

Tables are separate by model and test split. They report ordinary MSE as mean $\pm$ sample standard deviation with explicit $\times 10^m$ row scaling. Each variant is compared with `instance_mse` under MSE training and `instance_nmse` under nMSE training.

3 Interpretation

Last centering tests sensitivity to local level evolution; asinh tests sensitivity to tails and multiplicative scale. A gain is evidence for that specific preprocessing bias, not a general solution to conditional shift. Results should be examined across datasets because either transform may remove useful absolute-level information.